MAE 3127
Fluid Mechanics Laboratory

Prof. Kartik Bulusu, MAE Dept.

Topics covered today:

1. Reynolds Transport Theorem
2. Conservation of mass
3. Conservation of momentum

School of Engineering
& Applied Science
THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

Photo: Kartik Bulusu

Reynolds Transport Theorem:

V → Volume
E → Energy

} Extensive properties: Dependent on mass

v → volume per unit mass
e → energy per unit mass

} Intensive properties: Independent of mass

$$\boxed{V = \iiint v \rho \, d\mathcal{V}}$$

v → volume per unit mass OR specific volume

ρ → mass per unit volume

$$\boxed{E = \iiint e \rho \, d\mathcal{V}}$$

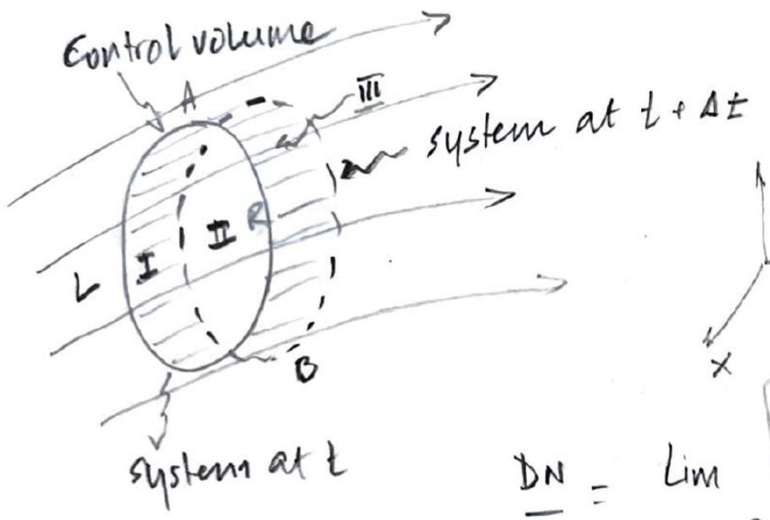
e → energy per unit mass OR specific energy

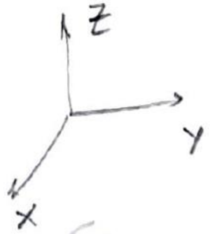
ρ → mass per unit volume

arbitrary extensive property

$$\boxed{N = \iiint \eta \rho \, dV}$$

arbitrary intensive property





$$\boxed{\left(\frac{dN}{dt}\right)_{system} = \frac{DN}{Dt}}$$

$$\frac{DN}{Dt} = \lim_{\Delta t \to 0} \left[\frac{\left(\iiint_{III} \eta \rho \, dV + \iiint_{II} \eta \rho \, dV \right)_{t+\Delta t} - \left(\iiint_{I} \eta \rho \, dV + \iiint_{II} \eta \rho \, dV \right)_{t}}{\Delta t} \right]$$

$$\boxed{\frac{DN}{Dt} = \oint_{CS} \eta \left(\rho \vec{V} \cdot d\vec{A} \right) + \frac{\partial}{\partial t} \iiint_{CV} \eta \rho \, dV}$$

Reynolds Transport Theorem

Conservation of Mass:

extensive property

velocity

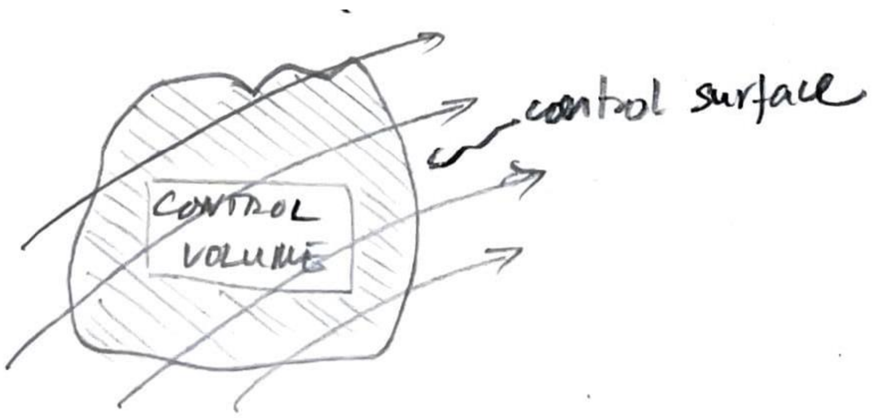
differential area

$$\boxed{\frac{DN}{Dt} = \oiint_{CS} \eta(\rho\vec{V}\cdot d\vec{A}) + \frac{\partial}{\partial t}\iiint_{CV} \eta\rho\, dV}$$

differential volume

Reynolds Transport Theorem

Intensive property OR specific property



Here: $N = M$ (i.e., mass of the fluid system)

η : specific mass $= 1$

$$\therefore \frac{DM}{Dt} = 0 = \oiint_{CS} \rho(\vec{V}\cdot d\vec{A}) + \frac{\partial}{\partial t}\iiint_{CV} \rho\, dV$$

$$\Rightarrow \boxed{-\frac{\partial}{\partial t}\iiint_{CV} \rho\, dV = \oiint_{CS} \rho(\vec{V}\cdot d\vec{A})}$$

(Rate of decrease of mass inside the control volume) = (Net efflux of mass through control surface)

Conservation of Linear Momentum :

Newton's Law :

$$\boxed{\vec{F}_R = \frac{d}{dt_{system}}\left[\iiint_M \vec{V}\,dm\right] = \frac{d\vec{P}}{dt_{system}}}$$

resultant external force acting on a system

time derivative is taken in the inertial frame of reference

velocity vector in inertial frame of reference

momentum vector

$$\boxed{\vec{F}_R = \frac{D}{Dt}\iiint_M \vec{V}\,dm = \frac{D\vec{P}}{Dt}}$$

Control volumes fixed in inertial space: Applying Reynolds Transport Theorem

Extensive property

velocity

differential area

differential volume

$$\boxed{\frac{DN}{Dt} = \oint_{CS} \eta(\rho \vec{v} \cdot d\vec{A}) + \frac{\partial}{\partial t}\iiint_{CV} \eta \rho \, dV}$$

Intensive or specific property

Reynolds Transport Theorem

$N = \vec{P}$ ⇝ linear momentum

$\eta = \dfrac{\vec{P}}{m}$ ⇝ specific momentum or momentum per unit mass

$\Rightarrow \eta = \vec{v}$

$$\therefore \boxed{\frac{D\vec{P}}{Dt} = \oint_{CS} \vec{v}(\rho \vec{v} \cdot d\vec{A}) + \frac{\partial}{\partial t}\iiint_{CV} \vec{v}(\rho \, dV)}$$

$$\therefore \boxed{\underbrace{\oint_{CS} \vec{T} d\vec{A}}_{\text{Total force distribution acting on the control surface}} + \underbrace{\iiint_{CV} \vec{B} \rho \, dV}_{\text{Total force distribution acting inside the control volume}} = \underbrace{\oint_{CS} \vec{v}(\rho \vec{v} \cdot d\vec{A})}_{\text{Rate of efflux of momentum across the control surface}} + \underbrace{\frac{\partial}{\partial t}\iiint_{CV} \vec{v}(\rho \, dV)}_{\text{Rate of increase of momentum inside the control volume}}}$$

Strategy to start solving problems:

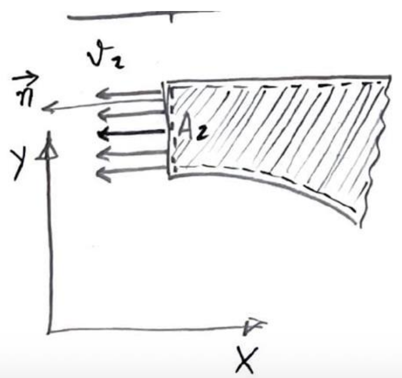
1. Set up a coordinate axis unless given to you.

Note: ① Orthogonal directions x, y & z are in the positive directions of the inertial frame of reference

② Direction of $(\vec{v} \cdot d\vec{A})$ is w.r.t the normal vector at the control surface

① & ② are independent of each other!

2. Choose a control volume in the region of interest.



$\begin{pmatrix}\text{Mass} \\ \text{flux}\end{pmatrix} = +\rho_2 v_2 A_2$ {∵ v_2 is in the same direction as $\vec{n}$}

In inertial frame of reference v_2 is in the opposite direction to the x-direction access

∴ $(-v_2)(\rho_2 v_2 A_2)$

3. State your assumptions that will help simplify the conservation equation.

4. Use the appropriate sign convention for the intensive property in the conservation equation.

5. Use the continuity or conservation of mass equation to simplify the problem further and solve!

Conservation of Linear momentum:

$$\boxed{\oiint_{CS} T_x \, dA + \iiint_{CV} B_x \rho \, dV = \oiint_{CS} v_x (\rho \vec{v} \cdot d\vec{A}) + \frac{\partial}{\partial t} \iiint_{CV} v_x \rho \, dV}$$

Assumptions:

1. Steady flow
2. Incompressible Flow
3. 1-D flow at inlet & outlets
4. Free jets at locations 2 & 3
5. Neglect any body forces

\# Unsteady term cancels out due to assumptions 1 & 2

\# Body force is neglected due to assumption 5

$\therefore$ We now have the following linear momentum eq.

$$\boxed{\oiint_{CS} T_x \, dA + R_x = \oiint_{CS} v_x (\rho \vec{v} \cdot d\vec{A})}$$